

DISTRIBUTION OF HEAVY METAL POLLUTANTS IN PUNNAKAYAL COASTAL SEDIMENTS, SOUTHEAST COAST OF INDIA

R. Sornaraj¹, J.S. Yogesh Kumar², R. Balasubramanian³ and S. Subburamen*

**Department of Ocean studies and Marine Biology, Pondicherry University, Port Blair, Andaman and Nicobar*

1. Research Department of Zoology, Kamaraj College, Thoothukudi-India

2. Zoological Survey of India, Andaman and Nicobar Regional Centre, Port Blair, Andaman and Nicobar Islands, India

3. People's Action for Development (PAD), Vembar, Thoothukudi, India

*E-mail: ssubburaman@rediffmail.com**

Abstract

This study is the first documented evidence on heavy metal levels in surface sediments of Punnakayal coastal environment in the Gulf of Mannar, Southeast coast of India. Sediment samples were collected every month from January to December, 2008. Chromium, Copper, Lead and Zinc metals concentrations were estimated from these samples. The concentrations of Chromium ranged between 15 – 40 ppm; Copper 18 – 48 ppm; Lead 15 – 36 ppm and Zinc 180 – 410 ppm in the surface sediment. Highest metal concentration is recorded in November, 2008 followed by May, 2008. The significant increase of metals in sediment is mainly due to pollution effect, related to anthropogenic wastes discharged through river run-off.

Keywords: Chromium, Copper, Lead, Zinc, Sediment.

1. Introduction

Sediments are the ultimate reservoir for the numerous potential chemical contaminants that may contain effluents originating from urban, agricultural and industrial lands and recreational activities (Apitz et al., 2005). Discharges from extensive urban and industrial development have adversely affected the quality of coastal water and sediments around the world (Tuncer, et al., 1998; Owen and Sandhu, 2000). In general, heavy metals tend to be adsorbed onto suspended particles and are scavenged from the water column into proximal sediment. Analysis of surficial sediment off Palos Verdes Peninsula, California, USA effectively demonstrated enrichment of heavy metals in an ocean out fall field (Hershelman, et al., 1977) and confirmed that the proximal sediment metal concentration was roughly five-fold higher than in sediments several kilometres from the pollution sources. Metals like mercury, cadmium, lead and nickel can have serious consequences if they are bioaccumulated by

organisms. The magnitude of the problems associated with diagnosis of Minamata (nervous disorder in human) and Itai-Itai disease (softening of bone) respectively due to mercury and cadmium poisoning is well documented. To avoid such deleterious effects on human and aquatic organisms by heavy metals, it is highly essential to assess the heavy metal concentration in coastal environment. Assessment of sediment quality is recognized as a critical step in estimating the risks associated with man-made pollution in riverine systems (Vignati et al., 2003).

There is no information available on trace metal concentrations for Punnakayal coastal environment, which receives agricultural drainage, industrial effluents, and untreated sewage outfall through Punnakayal and Athoor rivers respectively. The present work addressed the spatial distribution of heavy metals like Cr, Cu, Pb and Zn in surface sediments of Punnakayal coastal environment, Southeast coast of India.

*Corresponding Author

2. Geographical Location of the Sampling Station

Punnakayal estuary is closed to Punnakayal village and located 25 km south of Tuticorin, Tamil Nadu, between latitude $8^{\circ}37'$ to $8^{\circ}41'$ North and longitudes $78^{\circ}03'$ to $78^{\circ}09'$ East (Fig.1). Major ecosystem present in the vicinity is a patch of estuarine mangrove and rich coral biodiversity and forms one of the ecologically important biotopes, endowed with a unique kind of coastal environment in the Gulf of Mannar, South India as it retains connection with the Indian Ocean.

3. Methodology

Sediment samples were collected around Punnakayal coastal environment Southeast coast of India every month from January to December 2008. Sediments collected with the help of 5 cm internal diameter and 60cm long PVC pipe corer were oven dried. Heavy metal levels were analyzed from five replicates of sediment sample collected during each month. One gram of dried and powdered sediment sample was used for metal analysis by the hot acid digestion method. Sediments were digested by hot acid (conc. HNO_3 and HClO_3) as

PUNNAKAYAL STUDY AREA MAP

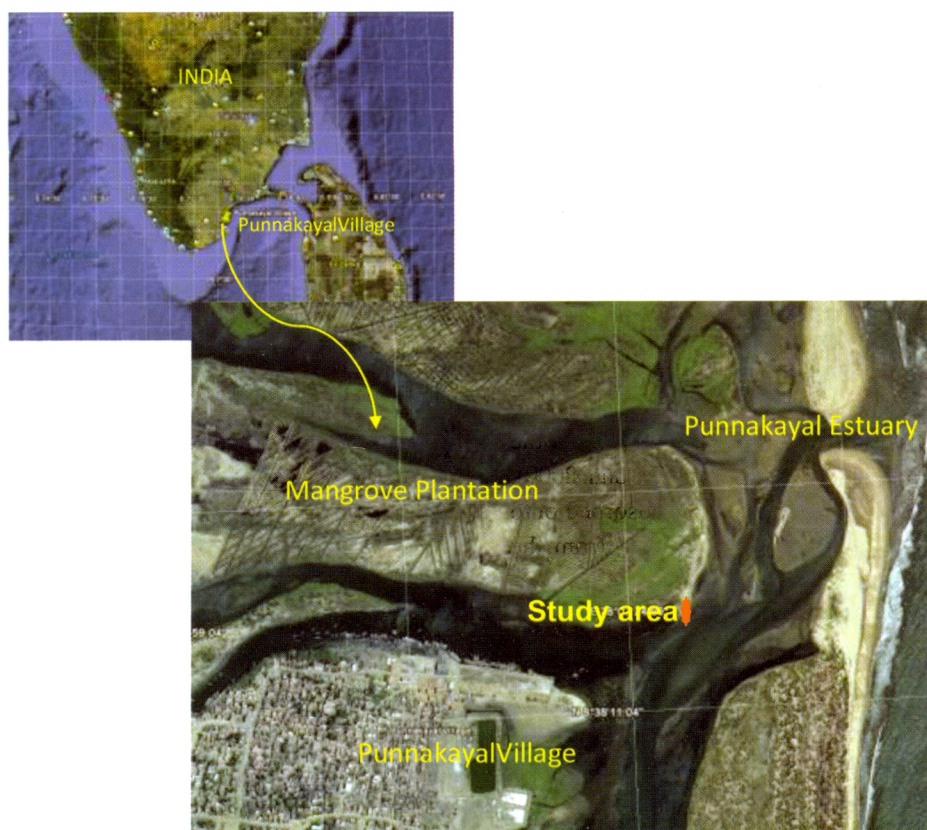


Fig.1. Satellite images of Punnakayal Estuary, Gulf of Mannar, India.

per the method of Chester and Hughes (1967). All the sediment metals concentrations were estimated using a Furnace Atomic Absorption Spectrometer, (Shimadzu Model AA 6800) with adequate number of standards for calibration of the instrument. The precision and detection limit of instrument is $\pm 0.01 \mu\text{g}$.

4. Results and Discussion

The concentrations of chromium in surface sediment ranged as 15 to 40 ppm; Copper 18 to 48

ppm; Lead 15 to 36 ppm and Zinc 180 to 410 ppm; in the surface sediment samples. Highest levels of metal concentrations were recorded during November 2008 by the end of northeast monsoon season and followed by May 2008. Lowest concentration of each metal was recorded in different months, which was not uniform with seasons. The analyzed sediment metal concentrations are shown in Figs. 2 to 5.

Concentration levels of sediment heavy metals recorded in the present study generally

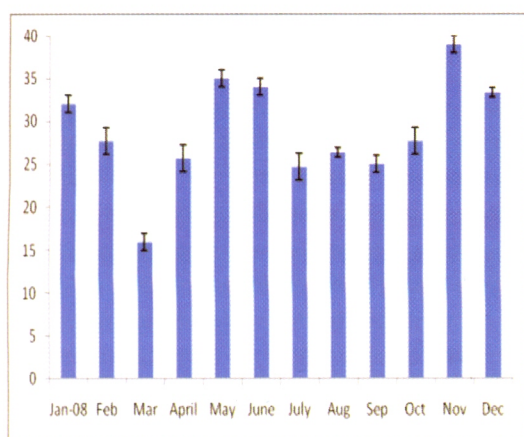


Fig. 2. Concentration of Chromium in Punnakayal coastal sediment.

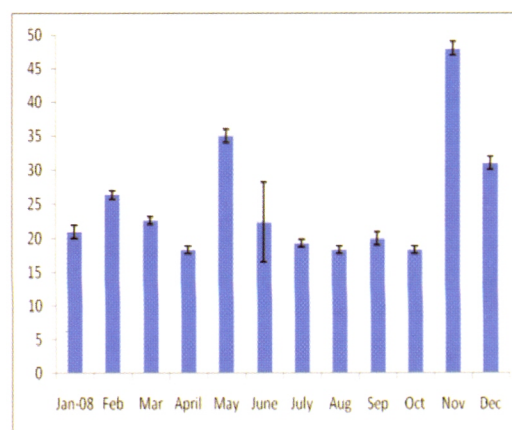


Fig. 3. Concentration of copper in Punnakayal coastal sediment.

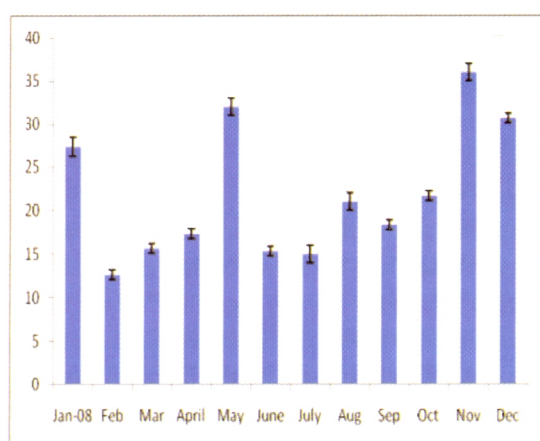


Fig. 4. Concentration of lead in Punnakayal coastal sediment.

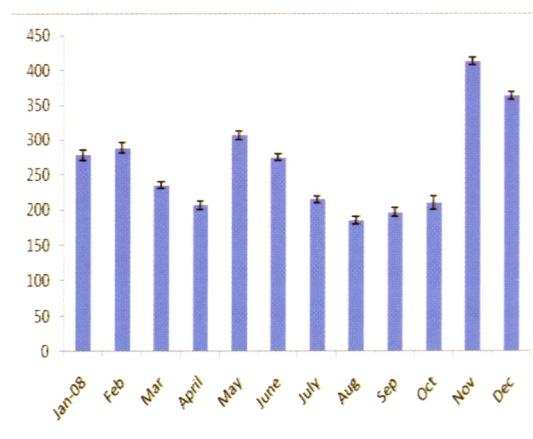


Fig. 5. Concentration of zinc in Punnakayal coastal sediment.

confirmed the earlier results obtained in similar environments. Chromium concentration levels in the current study are quite comparable to that reported by Donazzolo et al., (1984) and Owen and Sandhu (2000), who recorded 15 to 75 ppm and 14 to 30 ppm of chromium levels in the Adriatic Sea and Tolo harbour sediment samples respectively. Rajkumar and Persad (1994) recorded 15.95 ppm of Cu concentration in the sediments of near shore areas of Tobago, West Indies. These results are almost identical to the results obtained in the present investigation from Vaipar coastal sediment on Cu values.

Lead concentrations in Punnakayal coastal sediment samples are quite comparable to the study results reported elsewhere which ranged between 35 to 135 ppm in Tolo harbour, Hong Kong (Owens and Sandhu, 2000). Similarly Martin et al., (1983) recorded 9 to 438 ppm from Jakarta bay, Indonesia and 2.5 to 50.1 ppm of lead contents in the Taiwan coast, respectively.

In the current investigation the zinc concentrations were relatively high in all sediment samples. In the present study, the observed zinc concentration around Punnakayal coastal sediment samples are quite comparable to the results reported by John and Fanny (1984); Pragatheeswaran, et al., (1988) on sediment zinc contents in Greek (244 ppm), and Kodyakkarai coast (560 ppm), Southeast India, respectively. Donazzolo, et al., (1984) have also recorded similar zinc concentration (1.68 to 870 ppm) trend in the northern Adriatic Sea sediment samples.

Chromium, copper and lead input into the coastal environment is mainly through industrial and sewage effluents. Davide et al., (2003) study results showed that increased Cu, Cr and Pb concentration in Mediterranean rivers are mainly due to industrial and sewage direct outfall. Further Lopez-Sanchez et al., (1996) have also confirmed that the increased metal concentration off the coast of Barcelona is mainly due to sludge dumping. Tuncer et al., (1998)

estimated that 340 tons of Cu, 60 tons of Pb and 900 tons of Zn is annually discharged into the Turkish coast through river and stream runoff. They opined that land based activities are the main cause for dumping heavy metal contamination in Black Sea waters through sewage outfall. Elevated metal concentrations in Punnakayal coastal shore sediment samples are mainly influenced by river and stream runoff. This is further substantiated by the present record of highest levels of heavy metals during post monsoon season.

Hallberg (1974) noted a linear relationship between metal concentration and sediment organic carbon. Martin (1973) studied the fate of metals by flocculation and their subsequent deposition along with organic matter in the sediments. In the present investigation, concentrations of metals were found to be high in sediments during the post monsoon season. The increased levels of metals in sediment samples during post northeast monsoon and floods can be attributed to the settlement of metal ions along with organic matter on to the coastal bottom areas. However, no season wise uniform trend could be discerned in the current study on metal distribution trend in Punnakayal sediment.

Increased levels of metals in the present study mainly due to wastes from chemical industries very closed to our study area (SPIC, DCW and Thermal power station), sewage, land drainage and river run off floods into the Punnakayal coastal environment, which is more evident that during the year 2008 Punnakayal River was heavily flooded during both April and October months. The pollution problem around the Punnakayal coastal environment has been increasing over the last couple of years. However, the heavy metal concentrations have not yet reached alarming levels that could be detrimental to the biota of estuarine system. However, it is apparent that Punnakayal coastal sediment concentrations were highly influenced by river run off.

5. Conclusions

The metal concentrations in sediments from Punnakayal coastal environment, though low, are indicative of pollution from point sources in and around the coast and from diffuse sources such as urban runoff from near villages (Athure, Moukani and Punnakayal). Though the present results indicate that the studied metal concentrations were well within threshold limits, it is likely that the estuarine system will face heavy metal pollution since the surrounding area is facing heavy industrialization. This situation calls for regular monitoring so that the safe limits are not exceeded due to increased industrial activities. This may contribute to the degradation of the marine ecosystems in future.

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